CPSC 304 Project Cover Page

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

3. Our database is designed for a Fashion Magazine Management System, which models information revolving around fashion magazines and their associated entities, including publishers, articles, issues, contributors, and clothing. In the GUI, users can insert a new magazine or contributor, update magazine information, and delete magazines that don't have issues associated with them on or after a given cutoff date. Users can also search for clothing by type, fit, and/or designer, view selected contributor attributes, and find the first article of a given magazine. Additionally, users can display the number of authors per magazine, find publishers with above-average magazine counts for their country, identify the most prolific contributors by issue count, and find models who have worn every clothing item by at least one designer.

4. We applied our TA feedback from Milestone 2 and fixed the discrepancies in char/varchar types for some relations, deciding that the CREATE TABLE file we had would be our finalized relations. The following are the changed relations with the added missing candidate keys (Bolded means not null, underlined means primary key):

- Publisher_Location(address: VARCHAR2[30], city: VARCHAR2[20], **country**: char[2])
    - CKs: {address}

- Publisher(pname: VARCHAR2[20], country: char[2], address: VARCHAR2[30], established: Date)
    - CKs: {pname, country}
    - FKs: address (from Publisher_Location)

- Magazine(ISSN: char[8], mname: VARCHAR2[200], genre: VARCHAR2[20], language: VARCHAR2[20], pname: VARCHAR2[20], **country**: char[(2])
    - CK: {ISSN}, {mname, pname, country}
    - FK: pname and country (from Publisher)

- Issue(issuenum: int, ISSN: char[8], issuetitle: VARCHAR2[200], length: int, pdate: Date)
    - CK: {issuenum, ISSN}, {issuetitle, pdate, ISSN}
    - FK: ISSN (from Magazine)

- Article_PageRange(pagestart: int, pageend: int, pagecount: int)
    - CK: {pagestart, pageend}

- Article(doi: VARCHAR[220], header: VARCHAR2[200], pagestart: int, pageend: int, **issuenum**: int, **ISSN**: char[8])
    - CK: {doi}
    - FK: issuenum and ISSN (from Issue), pagestart and pageend (from Article_PageRange)

- Photo_FileType(filename: VARCHAR[220], filetype: char[5])
    - CK: {filename}

- Photo(pid: int, filename: VARCHAR2[20], filesize: int)
    - CK: {pid}
    - FK: filename (from Photo_FileType)

- Contains(doi: VARCHAR2[20], pid: int)
    -CK: {doi, pid}
    -FK: doi (from Article), pid (from Photo)

- Birthday_Age(birthday: date, age: int)
    -CK: {birthday}

- Contributor_EmailDetails(agency: VARCHAR2[20], cname: VARCHAR2[20], address: VARCHAR2[30], email: VARCHAR2[20], birthday: date)
    -CK: {email}
    -FK: birthday (from Birthday_Age)

- Contributor(sin: char[9], **email**: VARCHAR2[20])
    -CK: {sin}, {email}
    -FK: email (from Contributor_EmailDetails)

- Author(sin: char[9])
    -CK: {sin}
    -FK: sin (from Contributor)

- Photographer(sin: char[9])
    -CK: {sin}
    -FK: sin (from Contributor)

- Model(sin: char[9], height: int)
    -CK: {sin}
    -FK: sin (from Contributor)

- Photographed(sin: char[9], pid: int, camtype: VARCHAR2[20])
    -CK: {sin, pid}
    -FKs: sin (from Photographer), pid (from Photo)

- WorkedOn(sin: char[9], issuenum: int, ISSN: char[8])
    -CK: {sin, issuenum, ISSN}
    -FKs: sin (from Contributor), issuenum and ISSN, (from Issue)

- Clothes_Price(price: int, designer: VARCHAR2[20], ctype: VARCHAR2[20])
    -CK: {designer, ctype}

- Clothes(designer: VARCHAR2[20], ctype: VARCHAR2[20], cfit: VARCHAR2[20], cid: int)

-CK: {cid}
-FKs: designer and ctype (from Clothes_Price)

- AppearsIn(cid: int, issuenum: int, ISSN: char[8])
  -CK: {cid, issuenum, ISSN}
  -FKs: cid (from Clothes), issuenum and ISSN (from Issue)

- Wears(sin: char[9], cid: int, csize: char[20], fit: VARCHAR2[20])
  -CK: {sin, cid}
  -FKs: sin (from Model), cid (from Clothes)

- Brand(bname: VARCHAR2[20], established: date, country: char[2], VARCHAR2: char[20])
  -CK: {bname}

- Branded(bname: VARCHAR2[20], cid: int)
  -CK: {bname, cid}
  -FKs: bname (from Brand), cid (from Clothes)

5. The following are screenshots of the relations mocked up in a spreadsheet program (https://www.db-fiddle.com/):

Publisher_Location

```
676 SELECT * FROM Publisher_Location
```

Execute | Share | Oracle

ADDRESS	CITY	CO
1st St	New York	US
2nd St	Vancouver	CA
3rd St	London	UK
4th St	Paris	FR
5th St	Tokyo	JP

Publisher:

```
676 SELECT * FROM PUBLISHER
```

Execute | Share | Oracle

PNAME	CO	ADDRESS	ESTABLISHED
Vogue	US	1st St	01-JAN-12
Marie Claire	CA	2nd St	26-MAR-20
Elle	UK	3rd St	29-NOV-95
Glamour	FR	4th St	28-JUL-25
InStyle	JP	5th St	25-DEC-00

Magazine:

```
676 SELECT * FROM MAGAZINE
```

Execute | Share | Oracle

ISSN	MNAME	GENRE	LANGUAGE	PNAME	CO
12345678	The American Vogue	Fashion	English	Vogue	US
87654321	Trending Accessories	Trends	English	Marie Claire	CA
92096390	Elle Couture	Style	English	Elle	UK
82371382	La Semaine de Choissures	Shoes	French	Glamour	FR
10293847	InStyle Japan	Fashion	Japanese	InStyle	JP

Issue:

```
676 SELECT * FROM Issue
```

Execute | Share | Oracle

ISSUENUM	ISSN	ISSUETITLE	LENGTH	PDATE
1	12345678	January 2025 Issue	120	01-JAN-25
2	10293847	February 2025 Issue	110	01-FEB-25
1	87654321	March 2025 Accessories	95	01-MAR-25
1	92096390	Spring Couture	130	01-APR-25
1	82371382	Summer Shoes Special	80	01-JUN-26

Article_PageRange:

```
676 SELECT * FROM Article_PageRange
```

Execute | Share | Oracle

PAGESTART	PAGEEND	PAGECOUNT
1	5	5
6	10	5
11	15	5
16	20	5
21	25	5

Article:

```
676 SELECT * FROM Article
```

Execute | Share | Oracle

DOI	HEADER	PAGESTART	PAGEEND	ISSUENUM	ISSN
001A	Fashion Trends 2026	1	5	1	12345678
002B	Designer Spotlight	6	10	2	10293847
003C	Spring Accessories	11	15	1	87654321
004D	Couture Highlights	16	20	1	92096390
005E	Summer Shoes Must-Haves	21	25	1	82371382

Photo_FileType:

```
676 SELECT * FROM Photo_FileType
```

Execute | Share | Oracle

FILENAME	FILET
img001.jpg	JPEG
img002.png	PNG
img003.tif	TIFF
img004.bmp	BMP
img005.gif	GIF

Photo:

```
676 SELECT * FROM Photo
```

Execute | Share | Oracle

PID	FILENAME	FILESIZE
101	img002.png	2
102	img003.tif	3
103	img004.bmp	4
104	img005.gif	5
105	img001.jpg	1

Contains:

```
676 SELECT * FROM Contains
```

Execute | Share | Oracle

DOI	PID
001A	101
002B	102
003C	103
004D	104
005E	105

Birthday_Age:

```
676 SELECT * FROM Birthday_Age
```

Execute | Share | Oracle

BIRTHDAY	AGE
12-MAY-90	36
23-SEP-85	41
30-NOV-92	33
15-JAN-00	26
08-JUL-98	28

Contributor_EmailDetails:

```
676 SELECT * FROM Contributor_EmailDetails
```

Execute | Share | Oracle

EMAIL	AGENCY	CNAME	ADDRESS	BIRTHDAY
alice@gmail.com	FashionAgency	Alice	1st Ave	12-MAY-90
bob@gmail.com	StyleCorp	Bob	2nd Ave	23-SEP-85
carol@gmail.com	Trendsetters	Carol	3rd Ave	30-NOV-92
dave@gmail.com	PhotoWorld	Dave	4th Ave	15-JAN-00
eve@gmail.com	ModellingPro	Eve	5th Ave	08-JUL-98

Contributor:

```
676 SELECT * FROM Contributor
```

Execute | Share | Oracle

SIN	EMAIL
111111111	alice@gmail.com
222222222	bob@gmail.com
333333333	carol@gmail.com
444444444	dave@gmail.com
555555555	eve@gmail.com

Author:

```
676 SELECT * FROM Author
```

Execute | Share | Oracle

SIN
111111111
222222222
333333333
444444444
555555555

Photographer:

```
676 SELECT * FROM Photographer
```

Execute | Share | Oracle

SIN
666666666
777777777
888888888
999999999
123456789

Model:

```
676 SELECT * FROM Model
```

Execute | Share | Oracle

SIN	HEIGHT
987654321	175
876543210	180
765432109	178
654321098	172
543210987	182

Photographed:

```
676 SELECT * FROM Photographed
```

Execute | Share | Oracle

SIN	PID	CAMTYPE
777777777	101	DSLR
888888888	102	Mirrorless
999999999	103	Film
123456789	104	DSLR
666666666	105	Smartphone

WorkedOn:

```
676 SELECT * FROM WorkedOn
```

Execute | Share | Oracle

SIN	ISSUENUM	ISSN
111111111	1	12345678
222222222	2	10293847
333333333	1	87654321
444444444	1	92096390
555555555	1	82371382

Clothes_Price:

```
676 SELECT * FROM Clothes_Price
```

Execute | Share | Oracle

DESIGNER	CTYPE	PRICE
Gucci	Dress	1200
Prada	Shoes	800
Versace	Jacket	1500
Dior	Hat	400
Chanel	Bag	2000

Clothes:

```
676 SELECT * FROM Clothes
```

Execute | Share | Oracle

CID	CTYPE	CFIT	DESIGNER
201	Dress	Slim	Gucci
202	Shoes	Regular	Prada
203	Jacket	Loose	Versace
204	Hat	OneSize	Dior
205	Bag	Medium	Chanel

AppearsIn:

```
676 SELECT * FROM AppearsIn
```

Execute | Share | Oracle

CID	ISSUENUM	ISSN
201	1	12345678
202	2	10293847
203	1	87654321
204	1	92096390
205	1	82371382

Wears:

```
676 SELECT * FROM Wears
```

Execute | Share | Oracle

SIN	CID	CSIZE	FIT
987654321	201	M	Slim
876543210	202	L	Regular
765432109	203	XL	Loose
654321098	204	OneSize	OneSize
543210987	205	M	Medium

Brand:

```
676 SELECT * FROM Brand
```

Execute | Share | Oracle

BNAME	ESTABLISHED	CO	OWNER
Gucci	01-MAR-21	IT	Kering
Prada	02-MAY-13	IT	Prada Group
Versace	02-DEC-78	IT	Versace
Dior	16-DEC-46	FR	LVMH
Chanel	19-AUG-10	FR	Chanel

Branded:

```
676 SELECT * FROM Branded
```

Execute | Share | Oracle

BNAME	CID
Gucci	201
Prada	202
Versace	203
Dior	204
Chanel	205

5. All SQL statements used can be found in queries.sql inside the SQL folder AND appService.js where it is used in the code.

- Query 1 (Insert):
  - **For grading**: Inserting into Contributor_EmailDetails — Line 4-5 in queries.sql AND Line 596-646 in appService.js
  - Extra: Inserting into Magazine — Line 9-10 in queries.sql AND Line 650-668 in appService.js.
- Query 2 (Update):
  - Updating Magazine — Line 14-16 in queries.sql AND Line 685-699 in appService.js.
- Query 3 (Delete):
  - Deleting Magazine — Line 20-27 in queries.sql AND Line 704-723 in appService.js.
- Query 4 (Select):
  - Selecting from Clothes — Line 33-35 in queries.sql AND Line 737-778 in appService.js.
- Query 5 (Project):
  - Projecting on Contributor_EmailDetails — Line 41-42 in queries.sql AND Line 788-805 in appService.js.
- Query 6 (Join):
  - Joining on Issue and Article — Line 47-50 in queries.sql AND Line 809-821 in appService.js.
- Query 7 (Group By)
  - Grouping by WorkedOn — Line 56-59 in queries.sql AND Line 825-832 in appService.js.
- Query 8 (Having)
  - Having on Magazine — Line 65-76 in queries.sql AND Line 836-856 in appService.js.
- Query 9 (Nested Aggregation)
  - Nested Aggregation on WorkedOn — Line 81-91 in queries.sql AND Line 860-878 in appService.js.
- Query 10 (Division)

- Division on Wears, Contributor_EmailDetails, Clothes — Line 95-113 in queries.sql AND Line 882-908 in appService.js.

7. Query 7-10 Explanation:
Query 7:

```
-- GROUP BY (WorkedOn)
SELECT ISSN, COUNT(*)
FROM WorkedOn w, Author a
WHERE w.sin = a.sin
GROUP BY ISSN;
```

This query counts how many authors worked on each magazine issue, identified by its ISSN. It groups all `WorkedOn` records by issue and returns the total number of authors for each one.

Query 8:

```
-- HAVING (Magazine)
SELECT pname, country, COUNT(*)
FROM Magazine m2
GROUP BY m2.pname, m2.country
HAVING COUNT(*) >  (
   SELECT avg(sum)
   FROM (
       SELECT COUNT(*) AS sum
       FROM Magazine m1
       WHERE country = m1.country
       GROUP BY m1.pname
   )
);
```

This query finds publishers that publish more magazines than the average number of magazines published by publishers in the same country. It first counts how many magazines each publisher has in a country, then keeps only the publishers whose count is above that country's average.

Query 9:

```
-- NESTED AGGREGATION (Contributor + WorkedOn)
SELECT w.sin, COUNT(*) AS issue_count
```

```
            FROM WorkedOn w
            GROUP BY w.sin
            HAVING COUNT(*) = (
                SELECT MAX(issue_total)
                FROM (
                    SELECT COUNT(*) AS issue_total
                    FROM WorkedOn
                    GROUP BY sin
                )
            );
```

This query finds the contributors who have worked on the greatest number of magazine issues. It counts how many issues each contributor worked on and returns only those whose count matches the maximum.

Query 10:

```
-- DIVISION (Wears)

SELECT e.cname
   FROM Contributor_EmailDetails e
   JOIN Contributor c ON c.email = e.email
   JOIN Model m ON m.sin = c.sin
   WHERE EXISTS (
       SELECT DISTINCT cl1.designer
       FROM Clothes cl1
       WHERE NOT EXISTS (
           SELECT *
           FROM Clothes cl2
           WHERE cl2.designer = cl1.designer
           AND NOT EXISTS (
               SELECT *
               FROM Wears w
               WHERE w.sin = m.sin
               AND w.cid = cl2.cid
           )
       )
);
```

This query finds models who have worn every clothing item from at least one designer. In other words, it checks whether a model has worn all clothes made by a particular designer and returns the names of the models who satisfy that condition.

8. No AI tools were used.